Sapienza Università di Roma

Department of Computer, Control and Management Engineering

Neural Model Predictive Control on a Unicycle

Intelligent Control

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Todo list

1 Introduction

outline of the work

unicycle approach

Model Predictive Control (MPC) has emerged as a powerful control strategy in robotics due to its ability to handle constraints and optimize system behavior over a prediction horizon. When combined with neural networks, which can learn complex system dynamics, it creates a promising framework for controlling nonlinear systems.

This report presents the implementation and analysis of a Neural Model Predictive Control (NMPC) scheme applied to a unicycle robot model. The unicycle serves as an excellent first-test case due to its nonholonomic constraints and nonlinear dynamics, making it both challenging and representative of real-world robotic systems, but at the same time its low complexity makes it ideal for testing without too many complications.

Our approach integrates deep learning techniques to approximate the unicycle's dynamics, which are then utilized within an MPC framework to generate optimal control inputs. We explore the effectiveness of this method in trajectory tracking tasks, comparing it with traditional control approaches.

The following sections detail the theoretical foundations, implementation specifics, and experimental results of our neural MPC system. We also discuss the challenges encountered and potential improvements for future work.

2 Model Predictive Control

Model Predictive Control (MPC) is an optimal control technique where the actions are chosen solving an optimization problem, composed by a set of constraints and an objective function to be minimized over a finite horizon.

2.1 Cost function

Our objective is to minimize at each step the error with respect to the reference trajectory, both in traslational and orientational dimension, while also maintaining the module of the input actions as low as possible. A terminal state term is present too, where the traslational and orientational errors of the last horizon step are tried to be minimized. Thus, the cost function is composed by three terms:

1. **Spatial error**: at each step of the horizon, errors along x-axis, y-axis and error of orientation are summed together

$$J_{sp} = W_{e_x} e_x + W_{e_y} e_y + W_{e_\theta} e_\theta \tag{1}$$

where

$$
\begin{aligned}
e_x &= x - x_{ref} &\quad (2a)\\
e_y &= y - y_{ref} &\quad (2b)\\
e_\theta &= \theta - \theta_{ref} &\quad (2c)
\end{aligned}
$$

2. **Terminal state**: to make sure the final stage is a good one, we add a penalty on errors along x-axis, y-axis and error of orientation in the terminal state

$$J_{term} = W_{e_{x_{term}}} e_{x_{term}} + W_{e_{y_{term}}} e_{y_{term}} + W_{e_{\theta_{term}}} e_{\theta_{term}} \quad (3)$$

3. **Input**: it penalizes high values of input controls

$$J_u = W_v v + W_\omega \omega \quad (4)$$

3 Physical Informed Neural Network

In our implementation, we have used a Physical Informed Neural Network to predict the next state of the system, given the control input obtained from the Model Predictive Control.

losses

pipeline

image pinn

4 Results

what we have obtained

problems

pinn vs pinnc

5 Conclusions